MCO-1D ORNITHOPTER

Commissioning Manual — Revision 18 — July 2026

The complete guide from assembly through Discovery flights. Every step is numbered. Do not skip steps.

Aircraft	MCO-1D metachronal cascade ornithopter — SAR testbed
MTOW	3.448 kg (33.8 N) — measure actual after build
Stall speed	4.85 m/s (9.4 kt) out of ground effect
VNE	28.0 m/s (54.4 kt)
Freq limit	3.0 Hz hard cap — cannot be overridden
Wings	8 — four tandem pairs W1 (front) to W4 (rear)
Wing spans	W1=772, W2=749, W3=735, W4=712 mm semi (measure actual)
Root chord ext.	230-250 mm at bolt circle, tapered to 165 mm over 180 mm
Winglets	6-inch 4-slot, cant 75-80°, 35° aft sweep
Surface devices	Turbulator tape upper 33-58 mm, lower 119-132 mm; covert flaps at 71 mm and 111 mm chord positions
Motors	8 × GoolRC 5010-14 360 KV — one per wing
Controllers	8 × ODrive S1 — ONE BOARD PER MOTOR
Encoder	On-board absolute magnetic — cold-start position known immediately
Control	Joystick L/R=Roll, F/A=Pitch. Pedals=Yaw. Lever=Throttle.
Hover mode	V<3 m/s: lever→altitude rate (50%=hold ±5%, ±400 fpm full)
FW mode	V>5 m/s: standard fixed-wing conventions throughout
Blend zone	3-5 m/s: smooth automatic transition between modes
Mechanisms	15 optimizer-weighted: 5 roll, 5 pitch, 5 yaw
Pitot airspeed	Pixhawk MS4525DO — required for hover mode and all morphing
Health monitor	Real S1 FET+motor temps at 5 Hz; errors at 1 Hz
TC-01 gate	FREE_HOVER / STOL_FIRST / STOL_ONLY / REDESIGN
Wave Library	49,180 lines — 33 hypotheses, 180 variants, 164 GS cmds
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PART I — WHAT THIS AIRCRAFT IS

1. WHAT THE MCO-1D IS

Eight independently controlled wings. Four tandem pairs. One motor per wing. One ODrive S1 per motor. Fifty commands per second. The computer can change what any individual wing is doing within 20 milliseconds. The mission is Search and Rescue — threading through confined spaces, hovering in wind, operating at low altitude where rotor-wash from a conventional drone would endanger people on the ground.

2. METACHRONAL CASCADE

The wings do not all flap at the same time. W1 starts, then W2 begins a fraction of a cycle later, then W3, then W4. Each rearward pair captures energy from the wake of the pair ahead — the same principle that gave plesiosaurs 60% more thrust and 40% better efficiency from their hind flippers than from working alone.

3. PER-WING SENSING

Sensor	Source
Angle of Attack	MCP3208 #1 (SPI1 CE1) reads pitch servo feedback pot. Actual AoA per wing, 0.044° resolution, 50 Hz.
Sweep position	MCP3208 #2 (SPI1 CE0) reads sweep servo pot. 50 Hz.
Stroke position	ODrive S1 pos_estimate via CAN. Absolute — cold-start position known immediately, no index search.
Pitot airspeed	Pixhawk MS4525DO → msg.airspeed → V_ms. Used by hover mode (speed gating) and all morphing hypotheses. Never GPS ground speed.
FET temperature	S1 on-board thermistor → read_temperatures() at 5 Hz. Real measured value, not an estimate.
Motor temperature	S1 motor thermistor. Real measured value.
Motor current	S1 Iq_measured via CAN. Rope wear proxy.
Axis error state	S1 heartbeat error register at 1 Hz.

4. WING MORPHING PROGRAMME

State	Name	Key property
State A	Flapping	Normal. Lift and thrust from unsteady aerodynamics.
State B	Stopped neutral	Fixed-wing lift = ½ρV²S·CL. Zero motor power.
State C	Stowed aft	Sweep servo to 85-95° aft. Near-zero area and gust load.

Hyp	Name	Purpose
H-33	Variable wing area (18v)	Stow/stop any pair. Sweep trims pitch.
H-34	Dynamic sweep (16v)	±20-40° per stroke. Fwd downstroke, aft upstroke.
H-35	Thrust/lift separation (14v)	Stopped pairs = wings. Active = engines.
H-36	Loiter and ferry (8v)	3 pairs stopped. 1 pair at minimum freq. ~3× endurance.

5. ODRIVE S1 ARCHITECTURE

■ WARNING: 8 × ODrive S1 = 8 boards × 1 motor each. This is not 4 boards × 2 motors. Do not attempt to share brake resistors.

CAN node	Wing
Node 0	W1 port
Node 1	W1 starboard
Node 2	W2 port
Node 3	W2 starboard
Node 4	W3 port
Node 5	W3 starboard
Node 6	W4 port
Node 7	W4 starboard

wing_idx == node_id (1:1). arb_id = (node_id<<5)|cmd_id for all wings. On-board absolute encoder: position known at power-on, no index search needed. vel_limit=150 rad/s on all 8 boards — configure with mco_odrive_setup.py.

6. CONTROL SYSTEM — THREE MODES

■ WARNING: The lever follows fixed-wing convention throughout. Forward lever = more thrust. 50% = altitude hold. This is the same as every fixed-wing aircraft you have flown.

Mode	Lever	Joystick	Pedals
HOVER MODE V < 3 m/s pitot	LEVER → altitude rate 50%±5% = hold. >50% = climb. <50% = descent. Max ±400 fpm at 0% or 100%.	JOYSTICK → attitude hold Roll = bank angle ±30°. Pitch = translation ±15°. Release = hold current attitude.	PEDALS → yaw / heading hold Pedals centred 1 s = auto heading hold. Pedal input = yaw rate command.
BLEND ZONE 3-5 m/s pitot	Smooth automatic transition between hover and fixed-wing conventions. No pilot action required. Both modes partially active.	blend = (V–3)/2 (0=hover, 1=FW)	
FIXED-WING MODE V > 5 m/s pitot	LEVER → thrust 50% = cruise/altitude hold. >50% = climb. <50% = descend.	JOYSTICK → pitch and roll Fwd stick = nose down / accelerate. Rt stick = bank right.	PEDALS → yaw Identical to fixed-wing rudder pedals. Coordinated turns naturally.

7. MOTOR HEALTH MONITORING

Phase	Scope	What is checked
Pre-flight	5 S1 checks	Heartbeats, encoder magnets, temperatures near ambient, axis errors, current baseline at 5 rad/s.
In-flight	6 monitors / wing / 50 Hz	FET temp (warn 70°C, crit 90°C), motor temp (warn 60°C, crit 75°C), current (warn 18A, crit 22A), rope wear (>20% rise), port/stbd asymmetry (>25%), axis errors.
Post-flight	5 checks at shutdown	Peak temps, rope trend, cooling state, error states, next-flight GO/NO-GO. Saved to last_flight_health.json.

8. TC-01 HOVER FEASIBILITY GATE

TC-01 step 16 automatically evaluates the measured thrust curve and classifies the aircraft. TC-04 reads this classification and adapts its complete instruction set accordingly.

Classification	Condition	Consequence
FREE_HOVER	Thrust ≥ MTOW AND thermal margin ≥ 0.2 Hz	Proceed to TC-02. TC-04 includes free hover steps.
STOL_FIRST	Thrust ≥ MTOW but thermal margin tight	Proceed to TC-02. TC-04 does transition circuits first, brief hover only with thermal monitoring.
STOL_ONLY	Thrust 70-100% MTOW	Skip TC-02. TC-04 is pure forward-flight profile. Consider: reduce MTOW, increase wing area.
REDESIGN	Thrust < 70% MTOW	Programme halts. Aircraft cannot sustain flight. Reduce MTOW, increase wing area, or relax thermal constraints.

✓ The hover frequency is estimated at approximately 1.6 Hz for the current design. At the 3.0 Hz thermal ceiling, the aircraft generates approximately 3.4× MTOW. Build the TC-01 fixture and load cell to handle 115 N minimum.

9. SAFETY RULES

■ WARNING: 3.0 Hz frequency cap, 96° amplitude cap, and AoA stall protection cannot be overridden from any command.

■ WARNING: H-33 State C (stow) and H-35/36 two-pair configurations are gated on the TC-03 CT-J curve. Do not attempt before TC-03 completes.

■ WARNING: All aerodynamic decisions use pitot airspeed (V_ms = msg.airspeed). GPS ground speed is never used for any aerodynamic calculation.

✓ H-33/34/35/36 State B variants (stopped at neutral) are safe for TC-04 first flight because stopped wings still generate fixed-wing lift.

✓ Panel margin bar CRITICAL (red): motor at dangerous temperature or fault. Software has already reduced vel_limit on affected wings. Land promptly.

PART II — HARDWARE

10. WING ASSEMBLIES

CF rod cut lengths	W1=772 mm, W2=749 mm, W3=735 mm, W4=712 mm semi-span. Cut from 1 m stock. Off-cuts 228-288 mm — save as spares.
Root chord extension	230-250 mm at bolt circle. Tapers to 165 mm over 180 mm span. Leading edge sweeps forward ~15° at root.
Root fairing	Convex dome, half-ellipse profile, covers sector wheel mounting disk. Must clear fuselage at maximum 85-95° stow.
Winglet attachment	Rod tip. 6-inch 4-slot at 75-80° cant, 35° aft sweep. Bond sleeve ≥30 mm along rod.
Upper turbulator tape	33-58 mm from leading edge. 0.10-0.15 mm zigzag tape, full span.
Micro-VGs (optional)	58-69 mm. Height 0.5-0.8 mm, counter-rotating pairs, 5-8 mm spacing.
Covert flap 1	Hinge at 71 mm. Extends aft to 104 mm (33 mm chord). Rigid plate.
Covert flap 2	Hinge at 111 mm. Extends aft to 144 mm (33 mm chord). Rigid plate.
Lower turbulator tape	119-132 mm from leading edge. 0.10 mm tape, full span.
Sector wheel	100° arc mechanical. 7:1 gear ratio. Verify before assembly.
Rope drive	Independent per wing. No shared rope port/stbd. Inspect before every flight.

11-13. ODRIVE S1, ENCODER MAGNETS, BRAKE RESISTORS

See Section 5 (ODrive S1 Architecture) for node assignments. Encoder magnet installation procedure:

Step	Action	Expected / Notes
1	Unpower motor	Disconnect S1 board before touching motor.
2	Press magnet onto shaft	Flush with back face. Diametrically magnetised disc.
3	Align S1 encoder chip	Directly over magnet centre. Gap: 1-2 mm.
4	Verify field status	mco_odrive_setup.py --check-magnet → status 1=OK.

■ WARNING: Brake resistors: one 2Ω 50W per S1 board (included). Each board has its own DC bus. Sharing resistors will damage boards.

14. SERVOS AND ADC CHIPS

Pitch GPIO	W1p=18, W1s=23, W2p=24, W2s=25, W3p=12, W3s=16, W4p=20, W4s=21
Sweep GPIO	W1p=4, W1s=17, W2p=27, W2s=22, W3p=5, W3s=6, W4p=13, W4s=19
Pitch ADC	MCP3208 #1 SPI1 CE1 (GPIO7) — CH0-CH7 pitch feedback
Sweep ADC	MCP3208 #2 SPI1 CE0 (GPIO8) — CH0-CH7 sweep feedback
SPI1	CLK=GPIO11, MOSI=GPIO10, MISO=GPIO9. /boot: dtoverlay=spi1-2cs
Series resistors	1kΩ on each of 16 feedback lines (8 pitch + 8 sweep)

15. PITOT TUBE

■ WARNING: Pitot tube is not optional. Hover mode uses pitot airspeed to select control law (hover below 3 m/s, FW above 5 m/s). All morphing hypotheses H-33/34/35/36 use pitot airspeed. GPS ground speed is never used for any aerodynamic decision.

Sensor	Pixhawk MS4525DO digital airspeed
Mount	Forward fuselage or boom in clean undisturbed airflow
Connect	Pixhawk airspeed I2C port
Data path	Pixhawk → MAVLink VFR_HUD → mco_mavlink.py → sensors[V_ms]
Hover mode use	V_ms < 3.0 m/s → hover control law active
Morphing use	H-33 stow trigger, H-34 sweep schedule, H-35 role switch, H-36 free lift

16. LOAD CELL (BENCH ONLY)

Capacity	100 N minimum. Aircraft generates ~115 N at 3.0 Hz thermal ceiling.
HX711	DT=GPIO26, SCK=GPIO7 (no conflict with SPI1 CE1 — bench only)
Calibrate	python3 mco_loadcell.py --calibrate (1 kg mass required)
TC-01 use	Thrust curve: force vs frequency at 1.5, 2.0, 2.5, 3.0 Hz
TC-02 use	Hover detection: reads zero when lift = weight
TC-01 step 16	Reads max_thrust_N from thrust curve, computes feasibility gate

17-18. FLIGHT COMPUTER, SENSORS, POWER, RADIO, GPIO

Pin	Function
GPIO 4	DS18B20 OAT One-Wire
GPIO 5	W3p sweep
GPIO 6	W3s sweep
GPIO 7	SPI1 CE1 pitch ADC / HX711 SCK
GPIO 8	SPI1 CE0 sweep ADC
GPIO 9-11	SPI1 MISO/MOSI/CLK
GPIO 12	W3p pitch
GPIO 13	W4p sweep
GPIO 16	W3s pitch
GPIO 17	W1s sweep
GPIO 18	W1p pitch
GPIO 19	W4s sweep
GPIO 20	W4p pitch
GPIO 21	W4s pitch

GPIO 22	W2s sweep
GPIO 23	W1s pitch
GPIO 24	W2p pitch
GPIO 25	W2s pitch / CAN interrupt
GPIO 26	HX711 DT bench
GPIO 27	W2p sweep
I2C1 GPIO2/3	IMU 0x68 + baro 0x76
SPI0 CE0	CAN adapter MCP2515/2518FD
UART GPIO14/15	RC SBUS + Pixhawk MAVLink 921,600 baud

PART III — ASSEMBLY — 24 STEPS

■ WARNING: Do not connect any battery until Step 18. Do not power ODrives until mco_odrive_setup.py has run (Step 22).

Step	Action	Expected / Notes
1	Mount airframe	Rigid. No flex between motor mounts under load.
2	Mount 8 sector wheels	Rotate by hand. Full 100° arc smooth, no binding.
3	Mount 8 motors	Loctite 243 on all fasteners. Vibration isolation pads. Verify 7:1 gear ratio: 7 motor turns = 1 full sector arc.
4	Install 8 rope drives	One per wing, independent. No shared rope between port and stbd. Tension correctly — too tight causes high baseline current.
5	Mount 8 wings	Attach to sector wheel pivot. Swing full 96° without binding.
6	Install encoder magnets (before mounting S1 boards)	Press 1/4" magnet onto each motor shaft centre, flush with back face. Do this before the S1 board is in position so you can see the shaft.
7	Mount 8 ODrive S1 boards	One per wing. Vibration isolation. Align encoder chip 1-2 mm above magnet centre. Wire: node 0→W1p, 1→W1s, 2→W2p, 3→W2s, 4→W3p, 5→W3s, 6→W4p, 7→W4s.
8	Connect brake resistors	One 2Ω 50W per S1 board (included). Connect to brake terminal. DO NOT share between boards.
9	Wire CAN bus	Pi adapter → node 0 → 1 → 2 → 3 → 4 → 5 → 6 → 7. 120Ω terminator at Pi end AND at node 7 end. Both required.
10	Set pitch servo neutrals — BEFORE INSTALLING	Apply servo power only (no motor power). Command all pitch servos to 0° AoA. Attach horn at that position.
11	Set sweep servo neutrals and test stow	Command all sweep servos to 60° (neutral). Attach horn. Then command to 95° and measure actual angle (must reach 85° minimum for stow). Record actual stow angle in build measurements.
12	Install 8 pitch servos	One per wing. Route feedback wire back to ADC.
13	Install 8 sweep servos	One per wing. Verify linkage clears at stow position.
14	Wire pitch servo PWM	GPIO to each pitch servo signal: W1p=18, W1s=23, W2p=24, W2s=25, W3p=12, W3s=16, W4p=20, W4s=21.
15	Wire sweep servo PWM	GPIO: W1p=4, W1s=17, W2p=27, W2s=22, W3p=5, W3s=6, W4p=13, W4s=19.
16	Wire pitch servo feedback to MCP3208 #1	Each pitch feedback → 1kΩ → MCP3208 #1 SPI1 CE1. W1p→CH0, W1s→CH1, W2p→CH2, W2s→CH3, W3p→CH4, W3s→CH5, W4p→CH6, W4s→CH7.
17	Wire sweep servo feedback to MCP3208 #2	Each sweep feedback → 1kΩ → MCP3208 #2 SPI1 CE0. Same CH mapping.

18	Mount Pi 4 and sensors	Pi on vibration isolation neoprene. IMU and barometer to I2C1 (GPIO2 SDA, GPIO3 SCL). Lidar facing straight down, clear of wing arc.
19	Install OAT sensor and pitot tube	DS18B20 data → GPIO4, 4.7kΩ pull-up to 3.3V. Pitot → Pixhawk airspeed I2C port. Mount pitot forward of all wings in clean undisturbed airflow.
20	Route power cables — DO NOT CONNECT BATTERIES	Main battery → 30A fuse → 8 S1 boards (each has independent connection). Servo battery → dedicated servo rail (isolated). Isolated BEC 5V/3A → Pi only.
21	Install RC receiver and Pixhawk	SBUS output → Pi GPIO15 RXD. Bind to transmitter. Pixhawk TELEM2 → Pi UART 921,600 baud.
22	First power-on — Pi only (no motors, no servos)	Connect isolated Pi BEC. Pi boots. SSH in. Confirm: python3 --version → 3.8 or later.
23	Install software (Part IV)	Complete all 9 software steps before powering motors.
24	Apply wing surface devices	Turbulator tape upper 33-58 mm, lower 119-132 mm. Full span strips. Covert flaps: hinge at 71 mm (to 104 mm) and 111 mm (to 144 mm). Rigid plate with light torsional spring on each hinge.

PART IV — SOFTWARE INSTALLATION — 9 STEPS

Step 1 — Install Raspberry Pi OS Lite 64-bit

Use Raspberry Pi Imager on your laptop. Device: Pi 4. OS: Raspberry Pi OS Lite (64-bit). Advanced Options: hostname=mco-1d, enable SSH, set username/password, configure WiFi, set locale. Click Write.

Step 2 — /boot/config.txt additions

```
dtoverlay=mcp2515-can0,oscillator=8000000,interrupt=25
dtoverlay=spi1-2cs
dtoverlay=w1-gpio,gpiopin=4
dtparam=i2c_arm=on
dtparam=spi=on
```

Step 3 — Install Python packages

```
pip install pymavlink python-can pigpio spidev smbus --break-system-packages
sudo systemctl enable pigpiod && sudo systemctl start pigpiod
```

Step 4 — Copy all software files

```
scp mco_wave_library.py mco_main.py mco_odrive_interface.py \
mco_odrive_setup.py mco_loadcell.py mco_build_measurements.py \
mco_preflight_checks.py mco_rc_verify.py mco_vibration_check.py \
mco_test_cards.py mco_log_analysis.py mco_logger.py mco_mavlink.py \
mco_gs_server.py mco_panel.py mco_ground_station.py \
mco_preflight.py mco_servo_interface.py mco_hypothesis_tool.py \
mco_can_setup.sh mco_install.sh pi@mco-1d.local:/home/pi/
ssh pi@mco-1d.local
mkdir -p /home/pi/logs /home/pi/knowledge
bash /home/pi/mco_install.sh
```

Step 5 — CAN bus

```
sudo reboot && ssh pi@mco-1d.local
sudo bash /home/pi/mco_can_setup.sh
ip link show can0 # must show UP
```

Step 6 — Configure ODrive S1 boards (apply main battery power first)

```
python3 /home/pi/mco_odrive_setup.py
```

Configures all 8 nodes: pole_pairs=7, vel_limit=150 rad/s, VELOCITY mode, on-board encoder, brake resistor enabled.

Step 7 — Verify encoder magnets

```
python3 /home/pi/mco_odrive_setup.py --check-magnet
```

All 8 nodes must return status 1. Status 0 = magnet missing. Status 2 = too close. Status 3 = too far.

Step 8 — Motor calibration

```
python3 /home/pi/mco_odrive_setup.py --calibrate
```

Motor resistance and inductance only. No encoder calibration needed — absolute encoder is self-calibrating. Motor will spin briefly. Ensure wing is free to move.

Step 9 — Set initial cascade phases (after enabling motors)

```
# From Python on Pi:
from mco_odrive_interface import ODriveInterface
odi = ODriveInterface()
odi.enable_all()
odi.set_initial_phases([0, 45, 90, 135, 180, 225, 270, 315])
# W1p=0° W1s=45° W2p=90° W2s=135° W3p=180° W3s=225° W4p=270° W4s=315°
# Adjust per actual cascade phase offsets in wave library.
```

PART V — PRE-FIRST-FLIGHT SEQUENCE — 7 STEPS

Run these steps in order before TC-01. Do not skip any.

Step	Action	Expected / Notes
1	Load cell calibration	python3 mco_loadcell.py --calibrate Place 1.000 kg mass on cell when prompted. Saves to /home/pi/knowledge/loadcell_cal.json. REQUIRED before TC-01.
2	Build measurements	python3 mco_build_measurements.py Measure 15 physical dimensions with tape measure and scale. Records actual wing spans, chord, MTOW, CG, sweep range, stow angle. Patches wave library with real values. ~30 minutes.
3	ODrive S1 configuration	python3 mco_odrive_setup.py All 8 nodes. vel_limit=150. pole_pairs=7. VELOCITY mode. Saves to non-volatile memory.
4	Encoder magnet verification	python3 mco_odrive_setup.py --check-magnet All 8 nodes must return field_status = 1. Any other status: correct before proceeding.
5	RC channel verification	python3 mco_rc_verify.py Confirms: CH1=Roll (joystick L/R), CH2=Pitch (joystick F/A), CH3=Lever (forward=more thrust — FIXED-WING CONVENTION), CH4=Yaw (pedals L/R). Verifies lever direction explicitly. Saves to /home/pi/knowledge/rc_mapping.json.
6	Pre-flight checks — all 19	python3 mco_preflight_checks.py --bench Includes 5 S1-specific checks: check_s1_heartbeats: all 8 boards heartbeating check_s1_encoder_magnets: all 8 magnets status 1 check_s1_temperatures_preflight: within 20°C of ambient check_s1_axis_errors_preflight: all errors clear check_s1_current_baseline: rope/bearing health at 5 rad/s All 19 checks must return PASS or WARN. Any FAIL blocks arming.
7	Vibration baseline	python3 mco_vibration_check.py --quick IMU and SPI baseline before any motors run. Run full sweep (mco_vibration_check.py) after TC-01-06.

PART VI — COMMISSIONING TESTS

Four test cards in sequence. Never advance to the next until the current one fully passes. S1 health monitoring is active throughout all cards.

Card	Arrangement	Notes
TC-01	Bench (post)	16 steps. 10 auto, 6 semi-auto. Ends with feasibility gate.
TC-02	Same post, captive	8 steps. Fully automatic. Finds hover frequency. Run only if TC-01 returns FREE_HOVER or STOL_FIRST.
TC-03	Truck-mount	6 steps. Fully automatic. Maps CT-J curve.
TC-04	First free flight	7 steps. Pilot in command. Profile set by TC-01 gate.

TC-01 BENCH TEST — 16 STEPS

Start: tc01_start Confirm actions: tc01_confirm Monitor: tc01_status

Step	Action	Expected / Notes
01	Servo neutral — AUTO	All 16 servos to neutral. ADC reads. PASS: all within ±2°.
02	Servo range ±45° — AUTO	Commands ±45°. PASS: all reach commanded range within ±3°.
03	Pitch ADC 8 channels — AUTO	MCP3208 #1 all channels. PASS: all 8 readable.
04	Sweep ADC 8 channels — AUTO	MCP3208 #2 all channels. PASS: all 8 readable.
05	S1 8-node comms — AUTO	read_all_states() on all 8 nodes. PASS: all heartbeating. First S1 temperature baseline recorded.
06	Motor spin 10 rad/s — AUTO	All 8 nodes. S1 reads real temps and currents. Verify with IR thermometer: all motors below 60°C. After this step: run mco_vibration_check.py (full sweep) before step 07.
07	Wing flapping 0.5 Hz 30 s — AUTO	Monitor rope drives visually. S1 thermal monitoring active throughout.
08	Wing flapping 2.5 Hz 60 s — AUTO	S1 reads real FET and motor temps each tick. PASS: all below 60°C. GS: motor_health to monitor.
09	Pitch ADC calibration — SEMI-AUTO	Software commands -45°, 0°, +45°. Operator visually confirms wing position. tc01_confirm at each. PASS: all 8 pitch servos calibrated, error <1°.
10	Sweep ADC calibration — SEMI-AUTO	Same as step 09 for sweep servos.
11	Frequency ceiling — AUTO	Steps 2.0 to 3.0 Hz. S1 thermal monitoring. Records thermal ceiling (highest freq before temps exceed limit).
12	Static thrust measurement — SEMI-AUTO	Confirm load cell fixture attached. tc01_confirm. Software steps 1.5-3.0 Hz. HX711 reads force at each frequency. PASS: max thrust ≥ MTOW (33.8 N). Saves thrust curve.

13	RC channel mapping — SEMI-AUTO	Confirms: CH1=Roll (joystick L/R), CH2=Pitch (joystick F/A), CH3=Lever (forward=thrust), CH4=Yaw (pedals). Verifies lever direction (forward = more thrust).
14	Safety caps and fault detection — SEMI-AUTO	3.0 Hz cap and 96° amp cap tested. Operator disconnects W1p motor, confirms fault detection fires.
15	Port/stbd symmetry — AUTO	AoA symmetry via MCP3208. PASS: asymmetry <5%.
16	Export and HOVER FEASIBILITY GATE — AUTO	Computes: thrust/MTOW%, thermal ceiling, hover freq estimate, thermal margin. Classifies: FREE_HOVER / STOL_FIRST / STOL_ONLY / REDESIGN. Prints full assessment to console. TC-04 reads this classification. Saves to tc01_hover_feasibility dict in tc_results.json.

✓ After TC-01-06: run mco_vibration_check.py before TC-01-07. Monitor S1 temperatures throughout with: motor_health on GS.

✓ TC-01 step 16 output determines everything that follows. Read the assessment carefully. FREE_HOVER → proceed to TC-02. STOL_ONLY → skip TC-02, go to TC-03. REDESIGN → stop programme.

TC-02 CAPTIVE HOVER — 8 STEPS

Same post and load cell as TC-01. Fully automatic. Run only if TC-01 returned FREE_HOVER or STOL_FIRST. Start: tc02_start Monitor: tc02_status

Step	Action	Expected / Notes
01	Load cell check — AUTO	HX711 responding. Calibration loaded.
02	Lift-frequency curve — AUTO	1.5-3.0 Hz. Load cell reads at each step. Computes dL/df slope for descent controller.
03	Find hover frequency — AUTO	Steps from 1.0 Hz in 0.1 Hz increments. HOVER FREQUENCY = frequency at which load cell reads ≤1 N. This is the most critical measurement. Records hover_freq_hz.
04	Confirm hover — AUTO	Holds at hover frequency 5 s. Residual ≤2 N. PASS.
05	AoA sweep at hover freq — AUTO	AoA 8-22°. Builds actual CL-alpha curve from load cell.
06	H-26 side force classification — AUTO	Differential AoA applied. Reads IMU roll rate and lateral g. Classifies: ROLL or LATERAL_FORCE. If LATERAL_FORCE: yaw_h26 mechanism activated in yaw weight pool.
07	Calibrate all controllers — AUTO	Applies hover_freq_hz and dL/df to descent controller. Calibrates hover mode: hover_mode_calibrate(hover_freq_hz). Saves calibration constants to knowledge file.
08	5-minute captive discovery — AUTO	Discovery Mode on post. Exports knowledge.

Key output 1	hover_freq_hz — actual hover frequency at current MTOW
Key output 2	dL/df slope — for descent controller calibration

Key output 3	H-26 classification — ROLL or LATERAL_FORCE
Key output 4	Hover mode calibrated — hover_mode_calibrate() called automatically

TC-03 TRUCK FORWARD FLIGHT — 6 STEPS

■ WARNING: Pitot tube must be reading valid airspeed before TC-03. V_ms must show non-zero when truck is at speed. Check: hover_mode_status on GS — V_ms should show pitot reading.

Step	Action	Expected / Notes
01	Stationary baseline — AUTO	Aircraft at hover freq. Truck stationary. Records baseline.
02	Transition sweep — AUTO	GS shows target truck speed. Finds minimum freq at each speed. Hover mode activates/deactivates automatically as V_ms changes.
03	CT-J curve — AUTO	6 truck speeds. Records CT, lift, pitot airspeed, advance ratio J. REQUIRED before H-33 State C / H-35 2-pair / H-36 stow variants.
04	Speed envelope — AUTO	IMU stability at increasing truck speeds.
05	Descent simulation — AUTO	Freq below hover at cruise. Load cell reads tendency.
06	30-minute discovery at cruise — AUTO	morphing_speed_start. H-33/34/35 State B variants. Mechanism weights accumulate. After session: mechanism_weight_report.

TC-04 FIRST FREE FLIGHT — ADAPTIVE PROFILE

■ WARNING: TC-04 cannot start if TC-01 returned REDESIGN. FREE_HOVER and STOL_FIRST profiles require TC-02 calibration. Software checks all prerequisites automatically at tc04_start.

TC-04 automatically selects one of three instruction profiles based on the TC-01 feasibility gate result:

Profile	Steps	Notes
FREE_HOVER	Steps: hover attempts → stability hold → altitude hold → roll + convention verification → transition to forward flight → 5-minute Discovery Mode.	Hover mode is fully active. Lever 50% = altitude hold. Pedals centred 1s = auto heading hold. Right stick = bank right. Verify each axis before manoeuvres.
STOL_FIRST	Steps: STOL circuits (forward flight first) → brief hover test → altitude hold at cruise → roll response at cruise → speed range → Discovery Mode.	Brief hover only. Monitor S1 temps continuously. FET alert = return to forward flight immediately.
STOL_ONLY	Steps: STOL takeoff → climb and cruise → altitude hold at cruise → roll response → minimum speed exploration → forward-flight Discovery.	No hover attempted. Pure forward flight programme. morphing_speed_start after step 3.

✓ The first action in any TC-04 profile is the convention check: small lever advance above 50% should produce a frequency increase (climb). Small right roll stick should bank RIGHT. Right pedal should move nose RIGHT. Verify all four axes before any manoeuvre. Abort immediately if wrong.

PART VII — FLIGHT OPERATIONS

23. PRE-FLIGHT CHECKLIST

Step	Action	Expected / Notes
1	python3 mco_preflight_checks.py --flight	All 19 checks. Battery interlock: <80% = NO-GO for TC.
2	Check last flight health	cat /home/pi/knowledge/last_flight_health.json — read GO/NO-GO from post-flight.
3	Area and wind	30 m zone clear. Wind <15 kt normal ops. <5 kt for TC-04.
4	Panel check	All margin bars green. Motor health bar normal. Any CRITICAL = do not arm.
5	Lever position	LEVER MUST BE AT MINIMUM (aft) before arming. Arming interlock blocks if lever above minimum.
6	ARM	Press ARM. Verify: small lever advance → motor sounds. Small right roll → correct roll tendency.

24. NORMAL FLIGHT PROCEDURES

TAKEOFF (hover-capable)	From armed. Advance lever slowly from 50% toward 60%. Aircraft lifts. Lever 50% = altitude hold.
TAKEOFF (STOL)	From armed. Advance lever to 80-90%. Allow ground roll. Apply forward pitch stick. Rotate at 3-5 m/s.
HOVER	Lever at 50%. Roll and pitch sticks hold attitude. Pedals centred 1s = heading hold. Advance lever for climb, retard for descent.
TRANSITION	In hover: advance lever above 60%. Apply forward pitch. Aircraft accelerates. Above 5 m/s: FW mode auto-engages. Lever 50% continues to mean altitude hold.
FORWARD FLIGHT	Lever 50-65% cruise. Joystick for attitude. Pedals for coordinated turns.
RETURN TO HOVER	Reduce lever toward 50%. Retard pitch stick. Aircraft decelerates. Below 3 m/s: hover mode re-engages. Lever 50% = altitude hold.
LANDING	Hover above touchdown point. Lever below 50% = descent. 0% = maximum descent rate. Reduce lever smoothly to zero.

25. EMERGENCY PROCEDURES

Condition	Action
Panel: WARNING (amber)	Motor warm. S1 reducing vel_limit on affected wing(s). Continue with caution. Land at next opportunity. Check: motor_health on GS.
Panel: CRITICAL (red)	Motor at dangerous temperature or error. Land as soon as practicable. Software has idled affected wings. Remaining wings compensating.
Overcurrent alert	Rope binding or bearing failure. Reduce frequency. Land and inspect before next flight.

Rope asymmetry alert	Named wing drawing more current than its partner. Land. Inspect rope drive on named wing.
Loss of hover control	Advance lever above 50% for climb. Forward pitch for translation. Recover to forward flight (V>5 m/s) where FW mode is more stable.
h33_h34_h35_disengage_all	GS command. Returns all pairs to normal 4-pair flapping immediately.
Wing failure	7 wings: land when convenient. 6 wings: land within 2 min. 5 or fewer: land immediately.

26. POST-FLIGHT PROCEDURE

Step	Action	Expected / Notes
1	DISARM	All motors stop.
2	POST-FLIGHT button / lever to 0%	8-minute cooling period. S1 temperatures monitored automatically.
3	Post-flight health check (automatic)	check_s1_post_flight() runs at shutdown: peak temps, rope trend, cooling state, error states, GO/NO-GO. Saved to /home/pi/knowledge/last_flight_health.json.
4	Save mechanism weights	GS: save_mechanism_weights (also saves automatically at shutdown). mechanism_weight_report to see which mechanisms are converging.
5	Export and analysis	knowledge_export_usb and morphing_speed_report. On laptop: python3 mco_log_analysis.py for anomaly review.
6	Physical inspection	All 8 rope drives. Any fraying = replace before next flight. All 8 encoder magnets. Any loose = re-seat and verify.

PART VIII — CONTROL SYSTEM IN DETAIL

27. HOVER MODE (V < 3 m/s pitot airspeed)

Hover mode activates automatically when pitot airspeed drops below 3 m/s. No pilot action required. The mode switch is seamless.

Input	Function	Detail
LEVER	Altitude rate command	50% ± 5% = hold altitude automatically. >50% = climb. Rate scales from 0 at 55% to +400 fpm at 100%. <50% = descent. Rate scales from 0 at 45% to -400 fpm at 0%.
ROLL STICK	Bank angle command	Full right = bank +30°. Full left = bank -30°. Release = hold current bank angle (attitude hold).
PITCH STICK	Translation command	Forward = tilt nose forward to translate forward. Aft = tilt nose back to translate rearward. Release = hold current pitch attitude (attitude hold).
PEDALS	Yaw rate / heading hold	Right pedal = yaw nose right at up to 30°/s. Pedals centred for 1 second = heading hold engages automatically. Any pedal input cancels heading hold.

✓ Altitude hold is automatic whenever lever is within ±5% of 50%. This means 45% to 55% is the hold zone. Pulling lever toward 0% while in hold releases hold and begins descent.

28. FIXED-WING MODE (V > 5 m/s pitot airspeed)

Input	Function	Detail
LEVER	Thrust / altitude rate	50% = cruise / altitude hold. >50% = additional thrust / climb. <50% = reduced thrust / descent.
ROLL STICK	Bank angle	Right stick = bank right.
PITCH STICK	Pitch attitude / speed	Forward = nose down / accelerate. Aft = nose up / climb.
PEDALS	Yaw	Right pedal = nose right. Coordinated turns are natural in banked flight.

29. BLEND ZONE (3-5 m/s)

Between 3 and 5 m/s, the control law blends automatically. No pilot action. blend = (V – 3.0) / 2.0. At 3 m/s: fully hover. At 5 m/s: fully fixed-wing. Lever 50% = altitude hold throughout the blend zone — this does not change.

30. ALL 15 MECHANISMS

The pilot commands intent (roll, pitch, yaw). The software distributes that intent across all available mechanisms simultaneously, weighted by the optimizer-learned weights. All mechanisms run every tick.

Mechanism	Description
roll_amp	Amplitude differential port vs stbd. Fastest roll mechanism. Only in powered flight.
roll_sweep	Sweep asymmetry: inner wing swept aft. Both powered and glide.
roll_aoa	Per-wing AoA differential (H-25). 4× faster than amplitude. Always.

roll_phase	Cascade phase asymmetry port vs stbd. Powered flight only.
roll_dih	Dihedral asymmetry. Redirects lift. Always available.
pitch_freq	All wings flap faster or slower. Primary climb/descent in powered.
pitch_aoa	Collective AoA all 8 wings. Primary pitch in glide.
pitch_sweep	Collective sweep changes stroke plane tilt.
pitch_phase	Cascade phase advance biases thrust vector.
pitch_dih	Collective dihedral for flare and slow flight.
yaw_phase	Phase asymmetry port vs stbd. Primary yaw in powered flight.
yaw_dih	Dihedral asymmetry redirects thrust sideways.
yaw_amp	Amplitude asymmetry with opposing AoA for torque yaw.
yaw_sweep	Differential drag via asymmetric sweep.
yaw_h26	Lateral force (if TC-02 classifies H-26 = LATERAL_FORCE).

31. MECHANISM WEIGHT OPTIMIZER

Weights start at 0.200 each (equal). After every tick with meaningful pilot input, the IMU-measured rotation rate is compared to the expected rate. Mechanisms that were active when response was good gain weight. Mechanisms active when response was poor lose weight. Update rate: 0.002 per tick. Convergence: many flights.

Command	Purpose
mechanism_weight_report	Show current weight ranking for all 15 mechanisms.
save_mechanism_weights	Save to /home/pi/knowledge/mechanism_weights.json.
load_mechanism_weights	Load saved weights. Runs automatically at startup.
mechanism_weight_reset	Reset to equal (0.200) after hardware changes.
set_h26_classification LATERAL_FORCE	Activate yaw_h26 mechanism after TC-02-06 classifies the aircraft.

32. GLIDE AND UNPOWERED FLIGHT

Below approximately 10% lever, wings transition progressively to fixed-wing glide configuration. In pure glide (lever at 0%): all wings held at best-lift-to-drag AoA, zero motor power.

Axis	Glide control mechanism
Roll	sweep asymmetry (roll_sweep) and AoA differential (roll_aoa). Amplitude differential unavailable (no flapping). Roll by sweeping inner wing aft — identical to a bird gliding.
Pitch	Collective AoA (pitch_aoa) primary. More AoA = slower shallower glide. Less = faster steeper.
Yaw	Differential sweep drag (yaw_sweep) and dihedral (yaw_dih). Phase mechanism unavailable. Heading changes via banked turn.

PART IX — MOTOR HEALTH MONITORING

33. PRE-FLIGHT S1 CHECKS (5 checks in 19-check preflight)

Check	What it verifies
check_s1_heartbeats	All 8 boards must send CAN heartbeat. Missing = board not powered or CAN broken. Names the specific wing. FAIL blocks arming.
check_s1_encoder_magnets	Reads field_status from on-board encoder. 1=OK, 0=missing, 2=too close, 3=too far. Any non-1 status = named wing blocked from arming.
check_s1_temperatures_preflight	All motor and FET temps must be within 20°C of ambient (OAT sensor). Hot before flight = previous damage or insufficient cool-down.
check_s1_axis_errors_preflight	Reads error register all 8 boards. Auto-clears non-critical errors. Persistent errors = FAIL. Cleared errors = WARN (investigate cause).
check_s1_current_baseline	Spins all 8 motors at 5 rad/s for 2 s. High current at low speed = rope binding or bearing fault. P/S asymmetry >25% = uneven rope tension. Baseline stored for in-flight rope wear monitoring.

34. IN-FLIGHT S1 MONITOR (50 Hz, 6 monitors per wing)

Monitor	Threshold	Response
FET temperature	Warn 70°C / Crit 90°C	Above warn: vel_limit reduces proportionally via set_velocity_limit(). Above crit: motor idles (S1 hardware cutoff imminent).
Motor temperature	Warn 60°C / Crit 75°C	Above warn: panel WARNING. Land at opportunity. Above crit: panel CRITICAL. Land soon.
Motor current	Warn 18A / Crit 22A	Above warn: check rope and bearings after flight. Above crit: possible motor or drive fault.
Rope wear proxy	>20% current rise from baseline	Rising current at same freq = rope stretch, wear, or tension change. Named wing alerted. Inspect rope before next flight.
Port/stbd asymmetry	>25% within pair	One wing of a pair working harder than its partner. Inspect rope tension on higher-current side.
Axis errors	Any non-zero error code	Named wing with hex code. Auto-cleared if non-critical. Hard faults reported and require investigation.

Alert path: S1 hardware → read_temperatures/read_all_states at 5 Hz/1 Hz → S1InFlightMonitor.update() every tick → result[motor_health] → telem[motor_alerts] at 10 Hz → mco_panel.py margin bar.

35. POST-FLIGHT S1 CHECK (5 checks, automatic at shutdown)

Check	What it reports
Peak temperature review	Peak FET and motor temps vs thresholds. >75°C motor = inspect winding before next flight.
Rope trend (current rise)	>20% peak vs baseline = inspect named rope drive.
Post-flight temperature	Motors within 20°C of ambient = safe to store/cover.

Post-flight axis errors	Errors that developed in last moments. Investigate before next flight.
Next-flight readiness	GO / NO-GO summary. Saved to last_flight_health.json.

36. ALERT THRESHOLDS AND PILOT ACTION

Alert	Pilot Action
FET warn 70°C	vel_limit reduces. Continue with awareness.
FET crit 90°C	Motor idles. Land as soon as practicable.
Motor warn 60°C	Panel WARNING. Land at next opportunity.
Motor crit 75°C	Panel CRITICAL. Land soon.
Current warn 18A	Check rope after flight.
Current crit 22A	Motor or drive fault. Reduce frequency. Land.
>20% current rise	Rope wear. Inspect before next flight.
>25% P/S asymm	Rope tension uneven. Inspect.

PART X — WING MORPHING PROGRAMME

37. PITOT AIRSPEED REQUIREMENT

■ WARNING: All morphing decisions use pitot airspeed (V_ms = msg.airspeed from Pixhawk). GPS ground speed is never used for any aerodynamic calculation. Pitot also controls hover vs fixed-wing mode selection.

Free lift from stopped pairs at pitot airspeed (CL=1.2, best L/D AoA):

Pitot speed	1 pair stopped	2 pairs stopped	3 pairs stopped
6 m/s	1 pair: 7.5N (22%)	2 pairs: 15N (44%)	3 pairs: 19.7N (58%)
7 m/s	1 pair: 10.2N (30%)	2 pairs: 20.5N (61%)	3 pairs: 26.8N (79%)
8 m/s	1 pair: 13.3N (39%)	2 pairs: 26.9N (80%)	3 pairs: 35.0N (104%) ←
9 m/s	1 pair: 16.9N (50%)	2 pairs: 34.0N (101%) ←	3 pairs: 44.3N (131%)

← = stopped pairs cover full MTOW. Active pairs owe zero lift = pure thrust.

38. H-33 VARIABLE WING AREA (18 variants)

Type	Variants	TC gate
State B	Stopped at neutral (v09-v11)	Still generate fixed-wing lift. Safe TC-04.
State C, single pair	Stowed along fuselage (v01-v04)	Needs TC-03 CT-J.
State C, two pairs	v05-v08	Needs TC-03 CT-J.
Speed-adaptive	v12-v18	Auto-stow above threshold V_ms.

39. H-34 DYNAMIC SWEEP (16 variants)

Per-stroke sweep: forward on downstroke, aft on upstroke. 69% drag reduction on upstroke vs extended wing. ±20° variants (v01/v02) safe from TC-04. ±40° variants (v03/v06) gated on H-27 servo bandwidth data.

40. H-35 THRUST/LIFT SEPARATION (14 variants)

Configuration	Notes
Entry (TC-04 safe)	v05/v06: 1 lift wing, 3 thrust wings. Conservative.
2-pair separation	v01-v04: 2 lift, 2 thrust. Requires TC-03.
Speed-adaptive	v07: auto-selects role split by pitot airspeed.
Sprint	v13: 2 lift, 1 thrust, 1 stowed. Requires TC-03.

41. H-36 MINIMUM POWER LOITER AND FERRY (8 variants)

3 pairs stopped as fixed-wing lift generators. 1 pair at minimum frequency — pure thrust engine. Estimated ~3× endurance improvement vs normal cruise. v03 (W2 active) and v04 (W3 active) are best pitch-balanced — start there.

Variant	Notes

v03/v04	W2 or W3 active — best pitch balance. Safe TC-04.
v01/v02	W1 or W4 active — mild pitch tendency.
v06/v06b	Frequency hunting — finds minimum sustainable freq automatically.
v07	Ferry: maximise km per Wh. Needs stable forward flight at 6+ m/s.
v08	SAR loiter: minimum power priority. Objective: endurance × stability.

42. DISCOVERY MODES

Command	Purpose
morphing_speed_start	48-variant H-33/34/35 scheduler. Priority ordered: safe first, sprint last. 45 s per trial. Score: 45% speed + 35% gust + 20% efficiency.
loiter_endurance_start	8-variant H-36 scheduler. Score: 50% power + 30% alt stability + 20% roll.
disc_mode_engage	Full Discovery Mode. All 12 focus modes, all 33 hypotheses.
disc_mode_engage_focused gust_resilience	Gust rejection primary. Useful in turbulent conditions.

PART XI — REFERENCE

43. GPIO PIN REFERENCE

Pin	Function
GPIO 4	DS18B20 OAT
GPIO 5	W3p sweep
GPIO 6	W3s sweep
GPIO 7	SPI1 CE1 pitch ADC / HX711 SCK bench
GPIO 8	SPI1 CE0 sweep ADC
GPIO 9	SPI1 MISO
GPIO 10	SPI1 MOSI
GPIO 11	SPI1 CLK
GPIO 12	W3p pitch
GPIO 13	W4p sweep
GPIO 16	W3s pitch
GPIO 17	W1s sweep
GPIO 18	W1p pitch
GPIO 19	W4s sweep
GPIO 20	W4p pitch
GPIO 21	W4s pitch
GPIO 22	W2s sweep
GPIO 23	W1s pitch
GPIO 24	W2p pitch
GPIO 25	W2s pitch / CAN int
GPIO 26	HX711 DT bench
GPIO 27	W2p sweep
I2C1 GPIO2/3	IMU 0x68 + baro 0x76
SPI0 CE0	CAN adapter
UART GPIO14/15	RC SBUS + Pixhawk MAVLink

44. ODRIVE S1 NODE ASSIGNMENT

Node 0 = W1 port	Node 1 = W1 stbd
Node 2 = W2 port	Node 3 = W2 stbd
Node 4 = W3 port	Node 5 = W3 stbd
Node 6 = W4 port	Node 7 = W4 stbd

arb_id formula	(node_id << 5) | cmd_id — all wings, no exceptions
wing_idx == node_id	1:1 direct mapping. No axis0/axis1.
Cold-start position	Known immediately. No index search.
Brake resistors	One 2Ω 50W per board. Included. CANNOT share.

45. SOFTWARE FILE REFERENCE

File	Lines	Purpose
mco_wave_library.py	49,180	33 hyp, 180 variants, 164 GS cmds. Control layer. Hover mode. Morphing.
mco_odrive_interface.py	447	8×S1. node==wing. read_temperatures, read_all_states, set_velocity_limit.
mco_odrive_setup.py	282	--verify --calibrate --check-magnet.
mco_preflight_checks.py	1,256	19 pre-flight (5 S1). S1InFlightMonitor. check_s1_post_flight().
mco_test_cards.py	1,889	TC-01 step 16: feasibility gate. TC-04: 3 profiles.
mco_main.py	491	S1 temps 5 Hz. Errors 1 Hz. Dynamic vel_limit. Weight load/save.
mco_panel.py	506	Motor alerts on margin bar.
mco_rc_verify.py	351	CH1-4 map. Lever direction. Pedal direction.
mco_loadcell.py	527	HX711. Calibrate/tare/sweep/hover.
mco_build_measurements.py	328	15 measurements. Patches wave library.
mco_vibration_check.py	367	Sweep 0.5-3.0 Hz. SPI. Tick budget.
mco_log_analysis.py	462	Post-flight. Anomaly. Thrust curve.
mco_ground_station.py	660	motor_health, motor_peaks, hover_mode_status, mechanism_weight_report, morphing_speed_report.
mco_preflight.py	782	Three-tier preflight (original).
mco_logger.py	220	50 Hz CSV logger.
mco_mavlink.py	290	Pixhawk bridge. V_ms = msg.airspeed.
Other files	12,716	16 additional supporting files.

46. GROUND STATION COMMANDS (SELECTED)

Command	Function
arm / disarm	Arm or disarm all 8 wings
emergency_stop	All amplitudes to zero immediately
motor_health	Per-wing FET temp, motor temp, current, active alerts
motor_peaks	Worst values this flight
hover_mode_status	Blend, alt hold, heading hold, lever→fpm table

mechanism_weight_report	15-mechanism ranking for roll, pitch, yaw
save_mechanism_weights	Persist learned weights to disk
load_mechanism_weights	Restore saved weights
set_h26_classification ROLL/LATERAL_FORCE	After TC-02-06 result
tc01_start / tc01_confirm / tc01_status	TC-01 control
tc02_start / tc02_status	TC-02 captive hover
tc03_start / tc03_status	TC-03 truck flight
tc04_start / tc04_continue / tc04_status	TC-04 free flight
morphing_speed_start / morphing_speed_report	H-33/34/35 discovery
loiter_endurance_start / h36_status	H-36 loiter/ferry
h33_h34_h35_disengage_all	Return all pairs to normal flapping
disc_mode_engage	Full Discovery Mode
knowledge_export_usb	Save registry to USB
failure_briefing	Pre-flight failure performance table
sensing_display	Full per-wing AoA, sweep, stroke, effective AoA

47. CALIBRATION CONSTANTS PENDING TC-02

Constant	Current Estimate	Source
DESCENT_CTRL_HOVER_FREQ_HZ	~1.60 Hz	TC-02-03: load cell reads zero
DESCENT_CTRL_K_FREQ_PER_FPM	0.0009	TC-02-07: from slope calibration
hover_mode hover_freq_hz	~1.60 Hz	TC-02-07: hover_mode_calibrate() called auto
ACTIVE_DAMP_K_AMP	0.40	TC-04-05: roll response in free flight
H-26 classification	Unknown	TC-02-06: ROLL or LATERAL_FORCE
_H35_V_CRUISE_ENTRY	5.0 m/s	TC-03: first lift-wing transition
_H35_V_CRUISE	8.0 m/s	TC-03: 2-pair lift configuration

✓ The hover frequency estimate is ~1.60 Hz for the current wing geometry. TC-02 will find the actual value and calibrate all controllers automatically.

48. PRE-FIRST-FLIGHT CHECKLIST SUMMARY

Done	Item	Notes
✗	mco_loadcell.py --calibrate	With 1 kg mass. REQUIRED before TC-01.
✗	mco_build_measurements.py	15 physical measurements. REQUIRED before TC-01.
✗	mco_odrive_setup.py	Configure all 8 S1 nodes.
✗	mco_odrive_setup.py --check-magnet	All 8 nodes status 1.
✗	mco_odrive_setup.py --calibrate	Motor calibration (no encoder cal needed).
✗	mco_rc_verify.py	CH1=Roll, CH2=Pitch, CH3=Lever fwd=thrust, CH4=Yaw.
✗	mco_preflight_checks.py --bench	All 19 checks pass (inc. 5 S1).
✗	mco_vibration_check.py --quick	Baseline before first motor run.
✗	TC-01: tc01_start	16 steps. Read step 16 output carefully.
✗	mco_vibration_check.py (full)	After TC-01-06, before TC-01-07.
✗	mco_log_analysis.py --tc01	Review thrust curve and feasibility gate.
✗	TC-01 gate: FREE_HOVER?	If STOL_ONLY: skip TC-02. If REDESIGN: stop.
✗	TC-02: tc02_start	8 steps captive hover (if FREE_HOVER or STOL_FIRST).
✗	mco_log_analysis.py --tc02	Confirm hover frequency extracted.
✗	TC-03: tc03_start	6 steps truck forward flight.
✗	CT-J curve confirmed	Before H-33 State C / H-35 2-pair / H-36 stow.
✗	Pitot airspeed verified	V_ms non-zero at truck speed. Not GPS.
✗	TC-04: tc04_start	Profile adapts to TC-01 feasibility gate.

✗	TC-04 convention check	Before any manoeuvre: verify all 4 axes correct.
✗	mechanism_weight_report	After TC-04: check optimizer is learning.
✗	save_mechanism_weights	Save after every productive flight session.

ADDENDUM A — REVISION 18 CRITICAL REVIEW CORRECTIONS

The following corrections were made after a systematic programme-wide critical review. All implemented in current codebase. Full detail in prior revision notes.

ID	Change	Detail
A1	Pitot validity gate	Disconnected pitot no longer enables hover mode on ground. 3 consecutive VFR_HUD messages required. GS: pitot_status.
A2	Hover freq placeholder	Updated 2.50 Hz → 1.87 Hz (5.095 kg estimated mass). TC-02 overwrites with measured value.
A3	Actual MTOW in TC-01	TC-01 reads weighed mass from build_measurements.json. Falls back to design 3.448 kg with warning.
A4	TC-01 step 05b	Sector arc datum symmetry check before any freq > 0.5 Hz. GS: sector_arc_status.
A5	Rope baseline sentinel	NaN replaces None. Missing baseline → MONITORING_OFFLINE alert, not silence.
A6	Servo ADC calibration	ADC range 372-3723 counts for 0-170°. Programme all 16 servos to 170° max via AGF-SPV3.
A7	GC bounded pauses	gc.disable() at startup. gc.collect() every 100 ticks.
A8	Altitude floor	0.30 m AGL floor — descent blocked below this.
A9	Flutter advisory	TC-03 CT-J step: stopped wings — no flutter analysis. 1 m/s speed increments. IMU RMS step-increase = abort.
A10	Cascade phase variants	H-01 gains 30°/45°/60°/90° inter-pair variants for Discovery Mode.
A11	New GS commands	pitot_status, pitot_force_valid, sector_arc_status, flutter_status.

ADDENDUM B — STROKE SOFT-STOP SYSTEM

Cosine velocity taper in the final 10° before each stroke reversal. Reduces rope tension impulse at reversal by ~88%. ODrive vel_limit reduced inside zone to cap braking torque.

Zone depth	10° before each reversal (SOFT_STOP_ZONE_DEG)
Min velocity at reversal	35% of full speed (SOFT_STOP_MIN_VEL_FRAC)
Vel_limit in zone	55% of 150 rps = 82.5 rps
Rope impulse reduction	~88% (KE $\propto v^2 \rightarrow 0.35^2$ = 12% of full)
Thrust loss estimate	~5.6% (zone = 10.4% of half-stroke)
Position source	ODrive pos_estimate via CAN → sensors[motor_pos_rad]

TC-01 step 05c verifies zone entries accumulate at 0.5 Hz before frequency is raised. GS: soft_stop_status / soft_stop_enable / soft_stop_set_zone / soft_stop_set_min_vel.

ADDENDUM C — CG MANAGEMENT SYSTEM

Three integrated components: airborne detection, early-flight CG calibration, and stability margin scheduling.

C1. Airborne Detection

Three independent signals combined into a confidence score. Declared airborne when score ≥ 0.63.

Signal	Weight	Method
Height (lidar)	55%	0.0 at h < 0.20 m → 1.0 at h > 0.60 m
Climb rate	25%	0.6 if h_agl rising > 0.10 m/s for 3 samples
Freq vs hover	20%	Scales 0→0.5 as freq approaches hover_freq

CG calibration begins automatically at 1.5 m AGL for 3 continuous seconds. Runs 90 s at enhanced update rate (alpha=0.25). ARM resets detection state. GS: airborne_status.

C2. Stability Margin Scheduling

Effective static margin controlled by shifting lift centroid relative to physical CG via sweep bias. No mass movement required.

Intent	Margin	Effect
cruise / hover / loiter / ferry / thermal	+20 mm	Passive stability
climb / descend	+10 mm	Moderate stability
sprint	0 mm	Neutral — maximum responsiveness
dart / stoop / zoom_climb	-10 mm	Slight aft — agile
recover / emergency	+30 mm	Maximum passive stability

■ WARNING: Hard aerodynamic limit: -15 mm. Never command below this.

Transitions ramp at 10 mm/s. GS: cg_status / cg_calibrate_now / cg_reset / cg_set_margin.

C3. Lateral Fast Trim (swinging payload)

In hover (V < 3 m/s), persistent lateral_g is detected with a 0.5 s time constant. Three consecutive same-sign readings → differential dihedral correction applied, rate-limited at 5 mm/s. Oscillatory payload swing (sign alternates) does not trigger correction.

ADDENDUM D — PROGRESSIVE DAMAGE AND CONTROLLED LANDING

PILOT AUTHORITY IS PRESERVED THROUGHOUT ALL FAULT MODES. Roll, pitch, and yaw stick remain active at all times. Automation manages thrust and descent rate. The pilot steers to the landing zone.

D1. Automatic Mode Transitions

Mode	Trigger	Pilot authority
normal	All wings healthy	Full authority all axes
degraded	7 or 6 wings active	Trim biases + freq boost. Full pilot authority.
emergency	5 wings active	Advisory: land at next opportunity. Full authority.
land_now	≤ 4 wings active	Lever → descent rate. Stick steers. Auto-engaged.

D2. land_now Mode

Lever controls descent rate, not altitude. Roll/yaw/pitch stick steer to landing zone as normal.

Lever	Sink rate	Use
0%	0.50 m/s	Maximum safe descent
50%	0.20 m/s	Moderate approach
100%	0.08 m/s	Near-arrested (use near ground)

Terminal flare at 0.5 m AGL: frequency reduces to 65% of hover frequency over 0.5 s. Stick still active during flare. Touchdown detected at 0.08 m — all motors idle.

D3. Asymmetric Failure Attitude Control

AoA trim biases applied to surviving wings to cancel asymmetric thrust moment. Biases scale to zero at full stick deflection (≥ 0.80). Pilot has unimpeded authority at full deflection.

D4. Failure Performance Reference

Failure	Active	Mode	Effect
1 wing lost	7 active	Land at opportunity	No impact
2 wings lost	6 active	Land at opportunity	Slight roll tendency
3 wings lost	5 active	EMERGENCY — land soon	Reduced authority
4 wings lost	4 active	LAND NOW (auto)	Controlled descent
5+ wings lost	≤ 3 active	LAND NOW (auto)	Steering only

GS: controlled_landing_status / fault_wing_restore / failure_briefing

ADDENDUM E — FAA CERTIFICATION TOOLS (mco_certification_tools.py)

Five tools address certification gaps closable in software. All persist data to /home/pi/knowledge/certification/. GS: cert_status shows overall readiness.

E1. FMEA Generator

13 failure modes documented. No Catastrophic class. 6 Hazardous, 5 Major, 2 Minor. Two hardware gaps identified (F-007 dual flight computer, F-013 servo LiPo redundancy). GS: cert_fmea → fmea_report.json + fmea_report.csv

E2. Flutter Boundary Logger

Records IMU RMS vs pitot airspeed at each TC-03 truck speed step. Flutter onset defined at 1.5× baseline. Auto-records from TC-03 CT-J step. Required onset > VNE × 1.15 = 32.2 m/s. GS: cert_flutter_report / cert_flutter_config

E3. Structural Loads Recorder

Peak accel_z_g, lateral_g, rates, and motor current per flight. Warning at 60% of 3.5g limit (2.10g). End-of-flight summary on disarm. GS: cert_loads_report

E4. Rope Life Tracker

Miner's Rule $I^2 \cdot$cycles per wing. Warn at 80%, block arming at 100%. Soft-stop estimated life extension: ~8×. Wired into preflight runner. NOTE: Replace ROPE_LIFE_I2_CYCLES = 5.0e8 placeholder with physical rope fatigue test data before type certification. GS: cert_rope_report / cert_rope_replaced / cert_rope_check

E5. Handling Qualities Analyser + Optimizer Feedback

Post-processes flight log CSV for ADS-33E-PRF metrics: bandwidth (min 2.0 rad/s), phase delay (max 0.10 s), roll quickness (min 0.50), cross-coupling (max 0.25). Auto-runs on latest log at _shutdown(). Feeds back to mechanism weight optimizer: low bandwidth → boost roll_aoa/roll_phase; high cross-coupling → reduce pitch_sweep/pitch_dih. Level 1 streak ≥ 3 flights → raise _TARGET_RATES by 2°/s. GS: cert_hq_analyse / cert_hq_feedback_all / cert_hq_optimizer_status

End of Addenda — MCO-1D Commissioning Manual Revision 18